

Math Behind the 85% Ashflow Alpha Portfolio Allocation

IMC Prosperity Round 3 — Manual Trading Analysis

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1 Set-Up

The trading budget is

$$B = 1,000,000.$$

Let

$$x_i = \text{percentage of budget allocated to product } i,$$

where x_i is written in percentage points. For example, $x_i = 10$ means 10% of the budget. The dollar investment in product i is therefore

$$I_i = B \frac{x_i}{100}.$$

The fee used for the calculation was interpreted as a quadratic product-level fee:

$$F_i = B \left(\frac{x_i}{100} \right)^2.$$

So the total fee is

$$F = B \sum_i \left(\frac{x_i}{100} \right)^2.$$

This matters because the fee grows with the square of position size. Doubling one position doubles exposure, but quadruples the fee:

$$F(2x) = B \left(\frac{2x}{100} \right)^2 = 4B \left(\frac{x}{100} \right)^2.$$

2 Objective Function

Let a_i be the expected return edge for product i , after choosing the correct direction. For a buy, a_i is the expected positive return from going long. For a sell, a_i is the expected positive return from going short.

Using

$$y_i = \frac{x_i}{100},$$

the expected net PnL contribution from product i is approximately

$$\Pi_i = B(a_i y_i - y_i^2).$$

Therefore the portfolio objective is

$$\max_{y_i} \Pi = B \sum_i (a_i y_i - y_i^2),$$

subject to

$$\sum_i y_i \leq 1, \quad y_i \geq 0.$$

Unused budget has no direct value, but using extra budget is only good if the added expected return is larger than the extra fee.

3 Marginal Benefit Versus Marginal Fee

For one product,

$$\Pi_i = B(a_i y_i - y_i^2).$$

The marginal expected net PnL is

$$\frac{d\Pi_i}{dy_i} = B(a_i - 2y_i).$$

Adding more to product i is worthwhile only when

$$a_i > 2y_i.$$

At the unconstrained optimum,

$$a_i - 2y_i = 0, \quad \text{so} \quad y_i^* = \frac{a_i}{2}.$$

This is the key rule: a position should only be increased until its expected edge is roughly equal to its marginal fee penalty.

4 Why Not Force the Full 100%?

Suppose an extra allocation Δy is added to a product already sized at y_i . The extra net value is approximately

$$\Delta\Pi_i \approx B(a_i - 2y_i) \Delta y.$$

If the remaining signals are weak, uncertain, or already crowded, then a_i may be smaller than $2y_i$. In that case,

$$\Delta\Pi_i < 0,$$

meaning the extra allocation *lowers* expected PnL after fees.

That is why using 85% can be rational even though unused budget expires. The unused 15% is not being saved for value; it is being avoided because the expected marginal after-fee return is not strong enough.

5 Chosen Allocation and Fee Calculation

The proposed portfolio was:

Product	Direction	Allocation x_i	Fee contribution
Obsidian cutlery	Buy	8%	6,400
Pyroflex cells	Sell	12%	14,400
Thermalite core	Buy	13%	16,900
Lava cake	Sell	13%	16,900
Magma ink	Buy	9%	8,100
Scoria paste	Buy	5%	2,500
Ashes of the Phoenix	Sell	7%	4,900
Volcanic incense	Buy	4%	1,600
Sulfur reactor	Buy	14%	19,600
Total		85%	91,300

The total fee is

$$\begin{aligned} F &= 1,000,000 \cdot (0.08^2 + 0.12^2 + 0.13^2 + 0.13^2 + 0.09^2 + 0.05^2 + 0.07^2 + 0.04^2 + 0.14^2) \\ &= 91,300. \end{aligned}$$

6 Why the Allocation is Spread Across Products

The fee is convex. For a fixed total exposure, spreading across multiple good signals is cheaper than concentrating everything in one product.

For example, putting 20% into one product costs

$$1,000,000 \cdot (0.20)^2 = 40,000.$$

Splitting the same 20% equally across two products costs

$$1,000,000 \cdot (0.10^2 + 0.10^2) = 20,000.$$

So diversification has a direct fee advantage, but only across products where the news signal still gives positive expected edge.

7 Interpretation of the Selected Sizes

The strongest allocations went to signals with clearer one-day catalysts:

- **Sulfur reactor:** index inclusion creates likely mechanical buying pressure.
- **Thermalite core:** demand forecast sharply increases.
- **Lava cake:** health review and shelf removal are clearly negative.
- **Pyroflex cells:** the end of the tax cut is negative for demand.

Smaller allocations were used where the signal was weaker, more speculative, or more likely to be crowded:

- **Scoria paste:** bullish stockpiling narrative, but based on celebrity-style hype.
- **Volcanic incense:** social attention surge, but potentially already overextended.
- **Ashes of the Phoenix:** negative public backlash, but less directly tied to the tradable good's immediate price.

8 Final Conclusion

The 85% allocation was intentional. The decision comes from comparing expected marginal return against the quadratic fee penalty:

$$\text{Add more only if } a_i > 2y_i.$$

Because the last 15% of budget would likely need to go into weaker or more crowded signals, the expected after-fee value of using it was judged lower than the quadratic fee cost. Therefore, the portfolio uses the strongest signals while avoiding low-confidence marginal exposure.